CE 639: AI for Civil Engineering
Assignment 3

Task 1: Understanding the Two Problems

1.1 Facade Defect Classification (Image Task)

The Facade Building Defects (FBD) dataset contains 1591 images of building facades, each labeled into one of four defect categories: Crack, Spalling, Stain, and Vegetation. The task is a multi-class image classification problem, where the goal is to train a Convolutional Neural Network (CNN) to automatically identify the defect type present in a facade image.

This problem is important for automated structural inspection, enabling scalable monitoring of infrastructure without manual inspection.

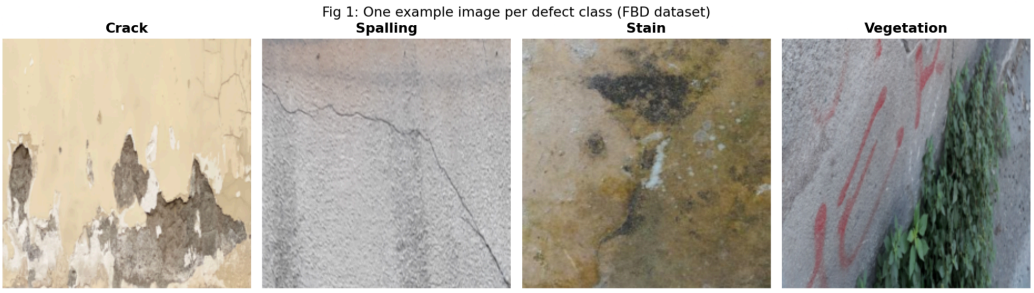

Fig: Samples of Dataset for all 4 Classes

1.2 Streamflow Forecasting (Time-Series Task)

The streamflow dataset contains hourly measurements of precipitation, evapotranspiration (ET), and discharge for a river catchment. The task is to predict the next-hour discharge using the past 24 hours of data.

This is a time-series regression problem, where the model must learn temporal dependencies between rainfall inputs and river discharge.

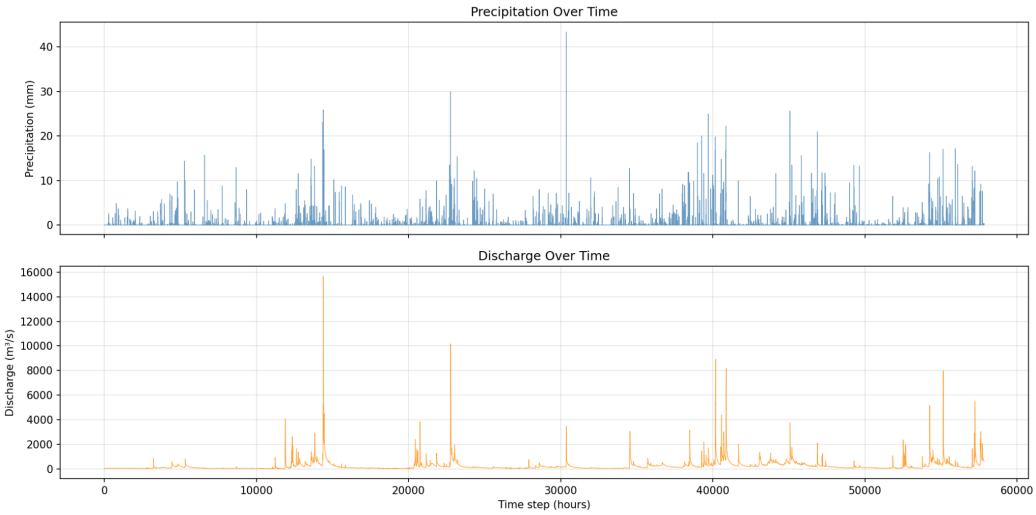

Fig: Prediction and Discharge time series

Task 2: CNN for Facade Defect Classification

2.1 Model Architecture

A simple 3-block CNN was implemented using PyTorch:

- **Block 1:** Conv(3→16) → BatchNorm → ReLU → MaxPool
- **Block 2:** Conv(16→32) → BatchNorm → ReLU → MaxPool
- **Block 3:** Conv(32→64) → BatchNorm → ReLU → MaxPool

The extracted features are flattened and passed through:

- Fully connected layer (256 neurons)
- Dropout (p = 0.5)
- Output layer (4 classes)

This architecture progressively extracts **low-level (edges)** to **high-level (defect patterns)** features.

2.2 Training Details

- Optimizer: Adam (lr = 0.001)

- Loss: CrossEntropyLoss
- Batch size: 64
- Epochs: 25
- Learning rate scheduler: StepLR

2.3 Results

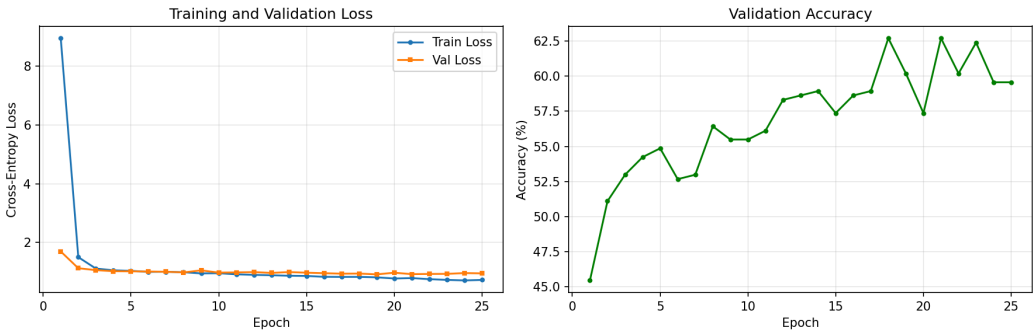

Fig: Training and validation Loss Curve

Final Validation Accuracy = 59.56%

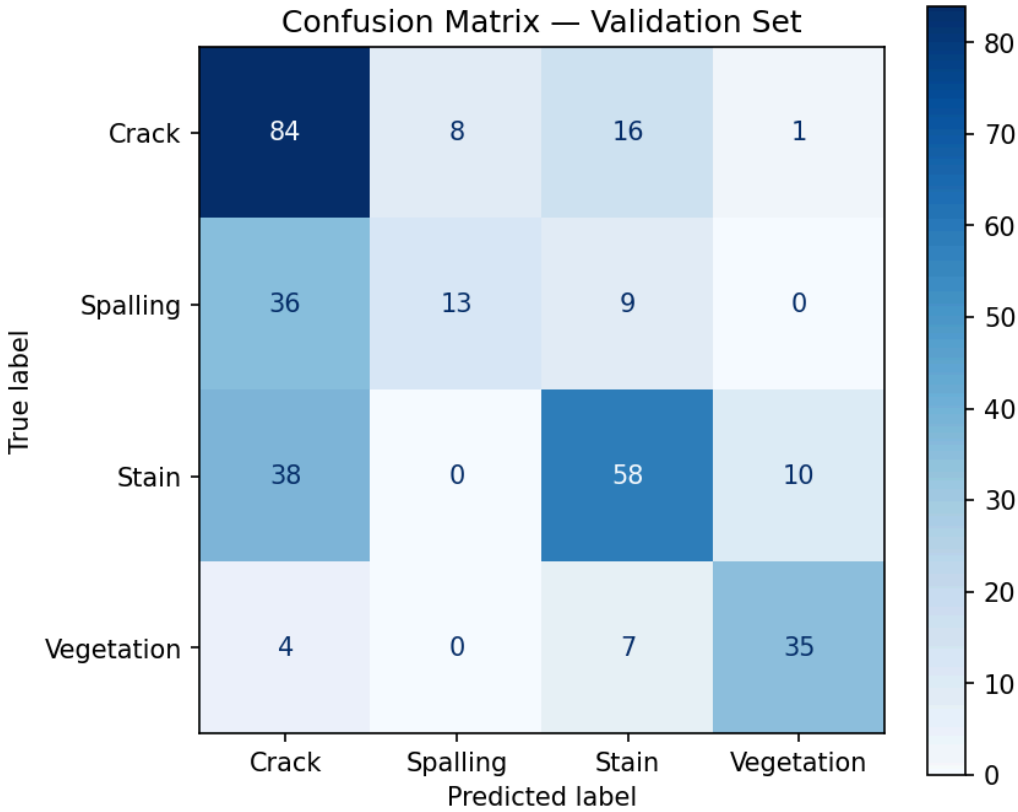

Fig: Confusion matrix

Class-wise Performance:-

Class	Precision	Recall	F1-Score
Crack	0.52	0.77	0.62
Spalling	0.62	0.22	0.33

Stain	0.64	0.55	0.59
Vegetation	0.76	0.76	0.76

Table : Class-wise precision, recall, and F1-score

Fig : Predicted and true labels

2.4 Observations

Easy and Difficult Classes

Easy classes: Vegetation tends to be easiest because it has a distinct visual signature green biological growth on grey/beige facades is highly separable from the other classes.

Difficult classes: Crack and Spalling are often confused because spalling involves surface detachment which can look like a large crack. Stain may also be confused with Spalling when discolouration accompanies surface detachment. The class imbalance (Crack has only 232 training images vs 436 for Spalling) further hurts Crack recall.

Task 3: Vanilla RNN for Streamflow Forecasting

3.1 Model Description

A vanilla RNN with:

- Hidden size: 64
- Input: 24-hour sequence (3 features)
- Output: next-hour discharge

The final hidden state is passed through a linear layer for prediction.

3.2 Results

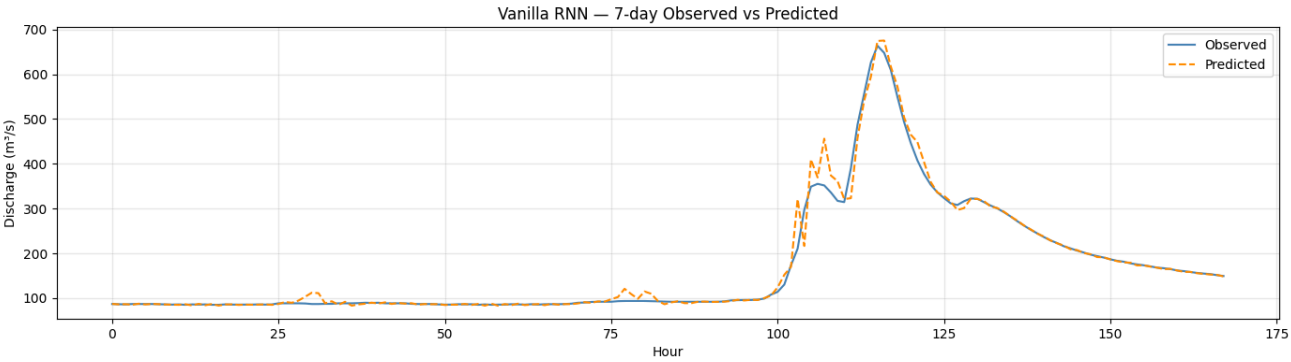

Fig : plot of observed vs predicted discharge for any continuous 7-day test period

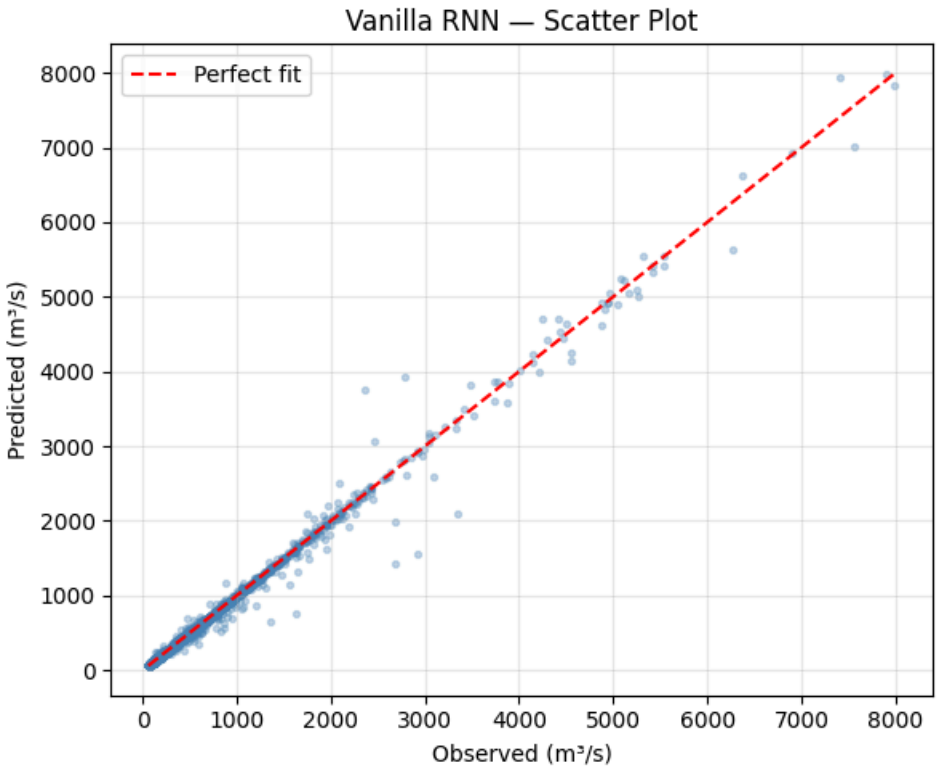

Fig : Scatter plot of predicted vs observed discharge

Metric	**Value**
MAE	9.2794 m^3/s
RMSE	43.6346 m^3/s
R^2	0.9922

Table : MAE, RMSE, and R 2 on the test set

3.3 Observations

The vanilla RNN struggles to capture peak discharge events. It tends to **underestimate flood peaks** due to the vanishing gradient problem, which limits its ability to retain long-term dependencies.

Task 4: LSTM for Streamflow Forecasting

4.1 Model Description

A single-layer LSTM with:

- Hidden size: 64
- Same input/output setup as RNN

The LSTM uses gating mechanisms (input, forget, output gates) to better retain long-term dependencies.

4.2 Results

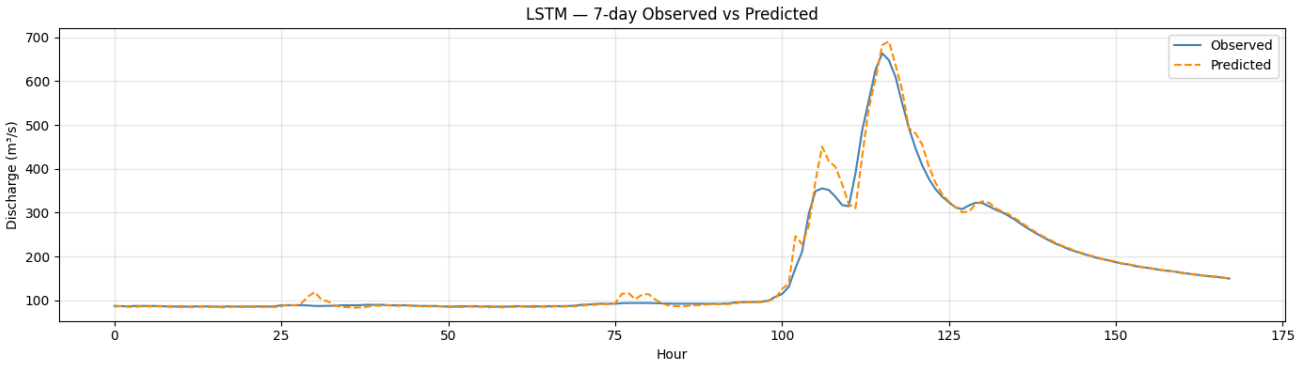

Fig : plot of observed vs predicted discharge for any continuous 7-day test period

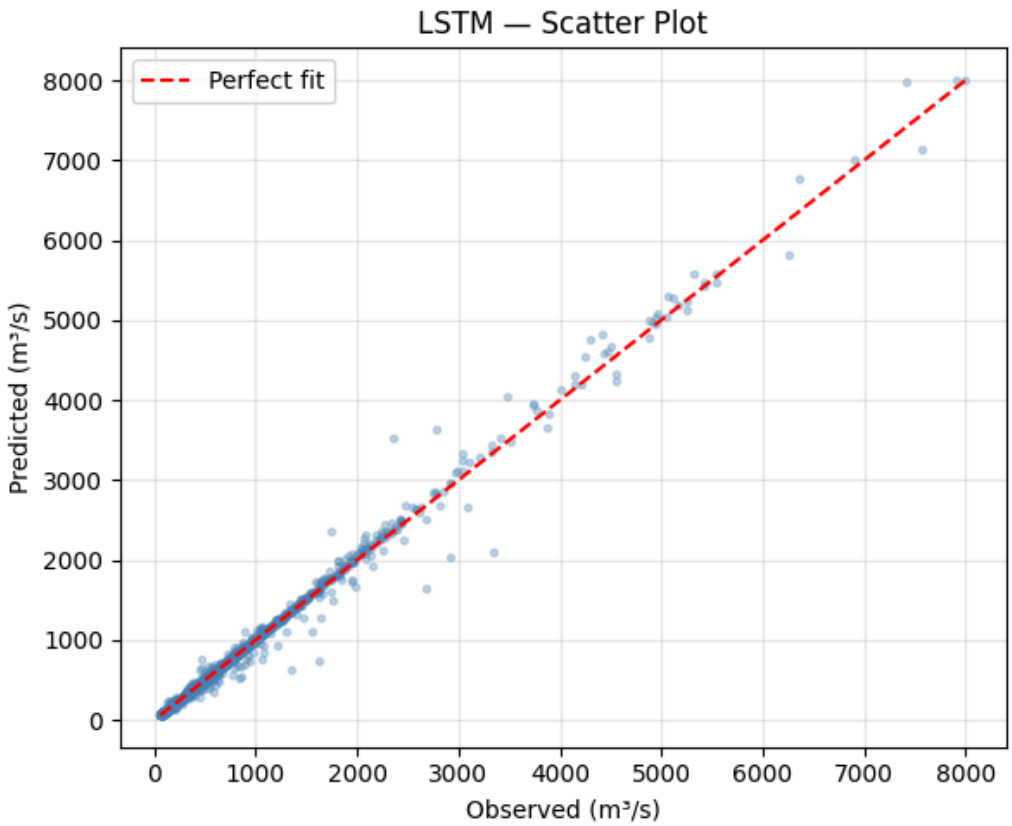

Fig : Scatter plot of predicted vs observed discharge

Metric	Value
MAE	7.7544 m³/s
RMSE	39.2755 m³/s

R^2	0.9937

Table : MAE, RMSE, and R 2 on the test set

4.3 Comparison with RNN

The LSTM performs significantly better than the vanilla RNN, especially during **peak discharge events**. Its gating mechanism helps retain information over longer time periods, allowing it to capture delayed rainfall-runoff relationships.

While both models perform similarly during low-flow periods, the LSTM provides **more accurate peak predictions** and overall lower error. However, both models still underestimate extreme flood peaks due to limited training examples.

Task 5: Short Discussion

Why is CNN suitable for the image task?

CNNs exploit spatial locality and translational invariance, making them ideal for detecting patterns such as cracks or stains regardless of their position in the image. Their hierarchical feature extraction allows learning from edges to complex defect structures.

Why are RNN/LSTM suitable for the streamflow task?

Streamflow is inherently sequential, depending on past rainfall and discharge. RNNs and LSTMs maintain temporal memory, enabling them to model lagged and nonlinear relationships in hydrological systems.

One limitation of the CNN

The CNN treats each image independently and ignores spatial context across larger structures. It is also sensitive to class imbalance and limited data, which may lead to biased predictions.

One limitation of sequence models

Both RNN and LSTM use a fixed 24-hour history, which may not capture longer hydrological memory such as groundwater flow. Additionally, models trained on a single catchment may not generalize to other regions.